KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 28 · MATH

Race Day in Boulder — Teacher Lesson Plan & Answer Key

Learning objectives: Students multiply tenths by tenths with a hundredths grid, multiply decimals to hundredths with an area model, partial products and estimation (counting decimal places to place the decimal point), divide by a decimal by multiplying the dividend and the divisor by the same power of 10, predict whether a product or quotient will be bigger or smaller than the starting number, and solve multistep pace, distance and money problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Lavinia (11), who runs on a youth cross-country team in Boulder, Colorado, at the foot of the Rocky Mountains. Her teammate Leandro loves the hills, Coach Larkin plans the workouts, Ms. Leighton, the head timer, records every finish time, and Lavinia's friend Lanelle helps at the water table. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs, extra spaces and a final period.
2. Slide 1 — Warm-Up, Part 1: Powers of 10 Review (4 pts, 2 each, Week 27 skill). If a student misses both warm-up slides, slide the digits on a place-value chart before moving on.
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): standard form, comparing powers of 10, dividing by 10^2, or a missing exponent.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W28.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Why Runners Count Decimals: nonfiction connection about Boulder, Colorado, at 5,430 feet (1,655 meters), with the Flatirons just west of town. Visitors from sea level are told to run more slowly at first because the air has less oxygen; elite runners train there, and the city hosts the BOLDERBoulder, one of the largest 10-kilometer road races in the United States. Cross-country races are measured in kilometers. No questions.
6. Slide 5 — Picture It: Tenths Times Tenths. On a hundredths grid, 3 columns (0.3) and 4 rows (0.4) overlap in 12 small squares, so 0.3 × 0.4 = 0.12 (3/10 × 4/10 = 12/100). Watch out: it is NOT 1.2; part of a part is smaller than both factors.
7. Slide 6 — Your Turn, Part 1: Tenths Times Tenths (8 pts, 4 each): a typed product and a choice (a product, or the squares shaded both ways on a grid).
8. Slide 7 — Your Turn, Part 2 (4 pts): multiply two tenths, then finish a second step (the part of the warm-up loop she walks, water before and after a race, or the area of two signs).
9. Slide 8 — Teach: Multiplying Larger Decimals. Coach Larkin's hill path is 1.5 times his 2.4-kilometer flat path. Area model: 2 + 0.4 + 1 + 0.2 = 3.6. Estimate 2 × 2 = 4; multiply 24 × 15 = 360; two decimal places in all make 3.60 = 3.6.
10. Slide 9 — Your Turn, Part 1: Multiplying Larger Decimals (8 pts, 4 each): a typed product and a choice (the best estimate, a whole-number fact, or the correct partial products).

11. Slide 10 — Your Turn, Part 2 (4 pts): estimate, multiply, then finish a second step (trail mix and dried apricots, laps plus a cooldown, or change from $10 for finish-line tape).
12. Slide 11 — Teach: Dividing by a Decimal. How many 0.4-kilometer sections fit in 2.8 kilometers? Multiply the dividend and the divisor by the same power of 10: 28 ÷ 4 = 7; 0.36 ÷ 0.04 = 36 ÷ 4 = 9. The quotient does not change, and dividing by a number less than 1 gives more than you started with.
13. Slide 12 — Your Turn, Part 1: Dividing by a Decimal (8 pts, 4 each): a typed quotient and a choice (the division with the same quotient, or the power of 10 that makes the divisor whole).
14. Slide 13 — Your Turn, Part 2 (4 pts): divide, then finish a second step (the difference between two runners' paces, full cups of water left, or rope pieces after the starting line is cut).
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (hundredth, partial product, estimate, decimal places, dividend, divisor, quotient, pace).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Bigger or Smaller? Predict Before You Compute. Multiplying by a decimal less than 1 makes the product smaller (8 × 0.5 = 4); dividing by a decimal less than 1 makes the quotient bigger (8 ÷ 0.5 = 16). Half of a 1.6-kilometer loop is 0.8 kilometer, but 1.6 ÷ 0.5 = 3.2 pieces.
18. Slide 17 — Your Turn: Bigger or Smaller? (6 pts, 3 each): type greater or less (or an estimate), and choose the answer that fits the prediction rule.
19. Slide 18 — Real World: Times and Paces (4 pts): a different situation in each set (Lavinia's 3-kilometer time used to predict a 5-kilometer race, Leandro's run-and-walk practice, or Lanelle's pace for 4 laps of a 0.75-kilometer loop).
20. Slide 19 — Real World: Racing at Altitude (4 pts): water before and after a race at 5,430 feet, jugs of sports drink from 1.2 kilograms of powder, or a map scale of 0.4 kilometer per centimeter.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a "which explanation is correct?" item and a "find the mistake" item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (quotation marks around dialogue, capital letters, commas with a city and state, and one grammar fix such as their/they're or to/too). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences (a series, a renaming phrase, or an opening word with a comma, and a verb that agrees with its real subject). Both serial-comma and no-serial-comma forms of a list are accepted.
24. Slide 23 — Match-Up Round (10 pts): 5 decimal products or quotients matched to their values, or division words (divisor, dividend, quotient, pace, hundredth) matched to meanings, 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick decimal questions against the student's own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: The Championship Meet (8 pts, 4 each): a different situation in each set (a 4-runner relay pace, an orange slice sale, or paint jars for a 1.8-meter by 0.6-meter team sign): choose the expression or first step, then finish the two-step answer.
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W28.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Race Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Boulder Decimal Champion!" with the congratulations message "Your race math passed every check in Boulder!" and a big confetti burst; below 85 shows "Good Work at the Meet!" with encouragement ("You are getting closer every time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: POWERS OF 10 REVIEW (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Multiplying by a power of 10 (Week 27): multiply 6.4×10^3. Product:	**6,400**	10^3 moves each digit 3 places to the left: 6.4 → 64 → 640 → 6,400.
r2_s0	Set 1: Exponents (Week 27): which is equal to 10 × 10 × 10 × 10? [10^4 / 10 × 4 / 4^{10} / 10^3]	**10^4**	Four tens are multiplied, so the base is 10 and the exponent is 4: 10^4 = 10,000.
r3_s0	Set 1: Standard form (Week 27): write $3 \times 10^3 + 4 \times 10^2$ in standard form. Standard form:	**3,400**	3 × 1,000 = 3,000 and 4 × 100 = 400. 3,000 + 400 = 3,400.
r4_s0	Set 1: Comparing (Week 27): 10^5 is how many times as large as 10^2? Times as large:	**1,000** (also accepted: 1,000 times)	10^5 has 3 more factors of 10 than 10^2, so it is 10 × 10 × 10 = 1,000 times as large.
r1_s1	Set 2: Dividing by a power of 10 (Week 27): divide $38 \div 10^2$. Quotient:	**0.38**	10^2 moves each digit 2 places to the right: 38 → 3.8 → 0.38.
r2_s1	Set 2: Multiplying by a power of 10 (Week 27): which product equals 5,200? [5.2×10^3 / 5.2×10^2 / 52×10^3 / 0.52×10^3]	**5.2×10^3**	5.2×10^3 = 5,200. The others: 5.2×10^2 = 520, 52×10^3 = 52,000, and 0.52×10^3 = 520.
r3_s1	Set 2: Dividing by a power of 10 (Week 27): divide $7.5 \div 10^2$. Quotient:	**0.075**	10^2 moves each digit 2 places to the right: 7.5 → 0.75 → 0.075. A placeholder zero holds the tenths place.
r4_s1	Set 2: Exponents (Week 27): 100,000 = 10 to what power? Type the exponent. Exponent:	**5** (also accepted: five)	100,000 has 5 zeros, so it equals 10^5. The exponent is 5.
r1_s2	Set 3: Multiplying by a power of 10 (Week 27): multiply 0.09×10^3. Product:	**90**	10^3 moves each digit 3 places to the left: 0.09 → 0.9 → 9 → 90.
r2_s2	Set 3: Dividing by a power of 10 (Week 27): which quotient equals 0.064? [$64 \div 10^3$ / $64 \div 10^2$ / $6.4 \div 10^3$ / $640 \div 10^3$]	**$64 \div 10^3$**	$64 \div 10^3$ moves each digit 3 places to the right: 0.064. The others: $64 \div 10^2$ = 0.64, $6.4 \div 10^3$ = 0.0064, and $640 \div 10^3$ = 0.64.
r3_s2	Set 3: Two steps (Week 27): evaluate $2.5 \times 10^2 - 4 \times 10$. Value:	**210**	2.5 × 100 = 250 and 4 × 10 = 40. Then 250 − 40 = 210.
r4_s2	Set 3: $27 \div 10^n = 0.27$. What is n? (Week 27) n =	**2** (also accepted: two)	The 7 moved from the ones place to the hundredths place, which is 2 places to the right, so n = 2.

YOUR TURN: TENTHS TIMES TENTHS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Multiply: 0.6 × 0.7 Product:	**0.42**	6 tenths × 7 tenths = 42 hundredths, so 0.6 × 0.7 = 0.42. On a grid, 6 columns and 7 rows overlap in 42 small squares.

ID	Question	Answer	Why (explain to student)
l2_s0	Set 1: Which product equals 0.08? [0.2 × 0.4 / 0.2 × 4 / 20 × 0.4 / 0.2 + 0.4]	**0.2 × 0.4**	2 tenths × 4 tenths = 8 hundredths = 0.08. The others: 0.2 × 4 = 0.8, 20 × 0.4 = 8, and 0.2 + 0.4 = 0.6.
l3_s0	Set 1: Two steps: Coach Larkin's warm-up loop is 0.8 kilometer. Lavinia jogs 0.6 of the loop and walks the rest. How many kilometers does she walk? Kilometers:	**0.32** (also accepted: 0.32 kilometers, 0.32 kilometer, 0.32 km)	Jog: 0.6 × 0.8 = 0.48 kilometer. Walk: 0.8 − 0.48 = 0.32 kilometer.
l1_s1	Set 2: Multiply: 0.9 × 0.3 Product:	**0.27**	9 tenths × 3 tenths = 27 hundredths, so 0.9 × 0.3 = 0.27.
l2_s1	Set 2: On a hundredths grid, Lavinia shades 0.4 one way and 0.6 the other way. How many small squares are shaded BOTH ways? [24 / 10 / 46 / 240]	**24**	4 columns and 6 rows overlap in 4 × 6 = 24 small squares, so 0.4 × 0.6 = 0.24.
l3_s1	Set 2: Two steps: a water bottle holds 0.9 liter. Leandro drinks 0.4 of a bottle before his race and 0.3 of a bottle after it. How many liters does he drink in all? Liters:	**0.63** (also accepted: 0.63 liters, 0.63 liter, 0.63 L)	Before: 0.4 × 0.9 = 0.36 liter. After: 0.3 × 0.9 = 0.27 liter. Total: 0.36 + 0.27 = 0.63 liter. (Or 0.7 × 0.9 = 0.63.)
l1_s2	Set 3: Multiply: 0.8 × 0.5 Product:	**0.4** (also accepted: 0.40)	8 tenths × 5 tenths = 40 hundredths = 0.40, which is the same as 0.4.
l2_s2	Set 3: Which number is equal to 0.7 × 0.2? [0.14 / 1.4 / 0.014 / 14]	**0.14**	7 tenths × 2 tenths = 14 hundredths = 0.14. The product is smaller than both factors.
l3_s2	Set 3: Two steps: one finish-line sign is 0.9 meter wide and 0.7 meter tall. A smaller sign is 0.5 meter wide and 0.4 meter tall. What is the total area of the two signs? Square meters:	**0.83** (also accepted: 0.83 square meters, 0.83 square meter, 0.83 sq m)	Big sign: 0.9 × 0.7 = 0.63 square meter. Small sign: 0.5 × 0.4 = 0.20 square meter. Total: 0.63 + 0.20 = 0.83 square meter.

YOUR TURN: MULTIPLYING LARGER DECIMALS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Multiply: 1.6 × 2.3 Product:	**3.68**	Estimate: 2 × 2 = 4. Multiply 16 × 23 = 368. Two decimal places in all, so the product is 3.68, which is close to 4.
k2_s0	Set 1: Which is the best estimate for 4.9 × 3.1? [15 / 1.5 / 150 / 7]	**15**	4.9 is about 5 and 3.1 is about 3, so the product is about 5 × 3 = 15. (The exact product is 15.19.)
k3_s0	Set 1: Two steps: for team snacks, Lavinia buys 2.5 pounds of trail mix at $4.80 per pound and 1.5 pounds of dried apricots at $6.20 per pound. What is the total cost? Dollars:	**21.30** (also accepted: 21.30 dollars, 21.30 dollar, 21.3, 21.3 dollars, 21.3 dollar)	Trail mix: 2.5 × 4.80 = $12.00. Apricots: 1.5 × 6.20 = $9.30. Total: 12.00 + 9.30 = $21.30.
k1_s1	Set 2: Multiply: 3.4 × 1.5 Product:	**5.1** (also accepted: 5.10)	Estimate: 3 × 2 = 6. Multiply 34 × 15 = 510. Two decimal places in all: 5.10, which is 5.1.
k2_s1	Set 2: 24 × 13 = 312. Use this fact to find 2.4 × 1.3. [3.12 / 31.2 / 0.312 / 312]	**3.12**	Each factor has one decimal place, so the product has two: 3.12. Check: 2 × 1 = 2 and 3 × 1 = 3, so the answer is between 2 and 4.

ID	Question	Answer	Why (explain to student)
k3_s1	Set 2: Two steps: at practice, Leandro runs 2.5 laps of a 1.2-kilometer loop. Then he jogs a 0.8-kilometer cooldown. How many kilometers does he cover in all? Kilometers:	**3.8** (also accepted: 3.8 kilometers, 3.8 kilometer, 3.8 km)	Laps: 2.5 × 1.2 = 3.0 kilometers. Add the cooldown: 3.0 + 0.8 = 3.8 kilometers.
k1_s2	Set 3: Multiply: 2.5 × 2.6 Product:	**6.5** (also accepted: 6.50)	Estimate: 3 × 3 = 9 is a little high, and 2 × 3 = 6 is a little low. Multiply 25 × 26 = 650. Two decimal places: 6.50, which is 6.5.
k2_s2	Set 3: Which partial products are correct for 3.2 × 1.4 in an area model? [3 + 1.2 + 0.2 + 0.08 / 3 + 12 + 2 + 8 / 3 + 0.12 + 0.2 + 0.8 / 3 + 0.4 + 0.2 + 0.08]	**3 + 1.2 + 0.2 + 0.08**	Split 3.2 into 3 and 0.2, and 1.4 into 1 and 0.4: 3 × 1 = 3, 3 × 0.4 = 1.2, 0.2 × 1 = 0.2, and 0.2 × 0.4 = 0.08. The sum is 4.48.
k3_s2	Set 3: Two steps: Ms. Leighton buys 3.5 meters of finish-line tape at $1.40 per meter. She pays with a $10 bill. How much change does she get? Dollars:	**5.10** (also accepted: 5.10 dollars, 5.10 dollar, 5.1, 5.1 dollars, 5.1 dollar)	Tape: 3.5 × 1.40 = $4.90. Change: 10.00 − 4.90 = $5.10.

YOUR TURN: DIVIDING BY A DECIMAL (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Divide: 3.6 ÷ 0.4 Quotient:	**9**	Multiply both by 10: 36 ÷ 4 = 9. Check: 9 × 0.4 = 3.6.
e2_s0	Set 1: Which division has the same quotient as 4.2 ÷ 0.7? [42 ÷ 7 / 42 ÷ 0.7 / 4.2 ÷ 7 / 420 ÷ 7]	**42 ÷ 7**	Multiply BOTH numbers by 10: 4.2 → 42 and 0.7 → 7. The quotient stays 6. The other choices change only one number, or multiply one number by 100.
e3_s0	Set 1: Two steps: Lavinia finishes a 3-kilometer race in 14.4 minutes. Leandro finishes the same race in 15.6 minutes. How many more minutes per kilometer is Leandro's pace than Lavinia's? Minutes:	**0.4** (also accepted: 0.4 minutes, 0.4 minute, 0.4 min)	Lavinia: 14.4 ÷ 3 = 4.8 minutes per kilometer. Leandro: 15.6 ÷ 3 = 5.2 minutes per kilometer. Difference: 5.2 − 4.8 = 0.4 minute.
e1_s1	Set 2: Divide: 0.48 ÷ 0.06 Quotient:	**8**	Multiply both by 100: 48 ÷ 6 = 8. Check: 8 × 0.06 = 0.48.
e2_s1	Set 2: Which division has the same quotient as 1.25 ÷ 0.5? [12.5 ÷ 5 / 125 ÷ 5 / 1.25 ÷ 5 / 12.5 ÷ 0.5]	**12.5 ÷ 5**	Multiply both by 10: 1.25 → 12.5 and 0.5 → 5. Both quotients are 2.5. The divisor only needs to be a whole number.
e3_s1	Set 2: Two steps: Coach Larkin has 4.5 liters of water. He fills cups that hold 0.3 liter each. If 12 runners each take one full cup, how many full cups are left? Cups:	**3** (also accepted: 3 cups, three)	Cups: 4.5 ÷ 0.3 = 45 ÷ 3 = 15 cups. Left: 15 − 12 = 3 cups.
e1_s2	Set 3: Divide: 8.4 ÷ 1.2 Quotient:	**7**	Multiply both by 10: 84 ÷ 12 = 7. Check: 7 × 1.2 = 8.4.
e2_s2	Set 3: To find 0.72 ÷ 0.08, Lavinia multiplies both numbers by the same power of 10. Which one makes the divisor a whole number? [100 / 10 / 8 / 0.1]	**100**	0.08 has two decimal places, so multiply both by 100: 72 ÷ 8 = 9. Multiplying by 10 gives 7.2 ÷ 0.8, and the divisor is still a decimal.

ID	Question	Answer	Why (explain to student)
e3_s2	Set 3: Two steps: Ms. Leighton has 7.2 meters of rope. She cuts off 1.2 meters for the starting line. Then she cuts the rest into pieces that are 0.6 meter long. How many pieces does she get? Pieces:	**10** (also accepted: 10 pieces, ten)	Rest: 7.2 − 1.2 = 6.0 meters. Pieces: 6.0 ÷ 0.6 = 60 ÷ 6 = 10 pieces.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: From the lesson: "A runner's pace is the number of minutes it takes to run one kilometer." Lanelle runs 2 kilometers in 9.4 minutes. What is her pace? [4.7 minutes per kilometer / 18.8 minutes per kilometer / 11.4 minutes per kilometer / 7.4 minutes per kilometer]	**4.7 minutes per kilometer**	Pace is minutes for ONE kilometer: 9.4 ÷ 2 = 4.7 minutes per kilometer.
v2_s0	Set 1: Type your answer: in 6.3 ÷ 0.9, the number 0.9 is called the ___. (one word)	**divisor** (also accepted: the divisor)	The divisor is the number you divide by. 6.3 is the dividend, and the answer, 7, is the quotient.
v1_s1	Set 2: From the lesson: "A tenth of a tenth is a hundredth, so 0.3 × 0.4 = 0.12." What is 0.1 × 0.1? [0.01 / 0.1 / 1 / 0.2]	**0.01**	A tenth of a tenth is a hundredth: 1/10 × 1/10 = 1/100 = 0.01.
v2_s1	Set 2: Type your answer: the answer to a division problem is called the ___. (one word)	**quotient** (also accepted: the quotient)	The quotient is the answer. In 2.8 ÷ 0.4 = 7, the quotient is 7.
v1_s2	Set 3: From the lesson: "An estimate tells you about how big the answer should be, so you know where the decimal point belongs." Which estimate helps you place the decimal point in 5.8 × 2.1? [6 × 2 = 12 / 58 × 21 = 1,218 / 5 + 2 = 7 / 0.6 × 0.2 = 0.12]	**6 × 2 = 12**	Round each factor to the nearest whole number: 6 × 2 = 12. So 5.8 × 2.1 must be about 12 (it is 12.18), not 121.8 or 1.218.
v2_s2	Set 3: Type your answer: in 6.3 ÷ 0.9, the number 6.3 is called the ___. (one word)	**dividend** (also accepted: the dividend)	The dividend is the number being divided. 0.9 is the divisor.

YOUR TURN: BIGGER OR SMALLER? (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Without computing: is 6.5 × 0.8 greater than or less than 6.5? Type greater or less.	**less** (also accepted: less than, less than 6.5, smaller)	0.8 is less than 1, so 6.5 × 0.8 is only part of 6.5. The product (5.2) is less than 6.5.
t2_s0	Set 1: Which answer is GREATER than 12? [12 ÷ 0.6 / 12 ÷ 1.5 / 12 ÷ 3 / 12 × 0.6]	**12 ÷ 0.6**	Dividing by a number less than 1 makes a bigger answer: 12 ÷ 0.6 = 20. The others: 12 ÷ 1.5 = 8, 12 ÷ 3 = 4, and 12 × 0.6 = 7.2.
t1_s1	Set 2: Without computing: is 4.2 ÷ 0.7 greater than or less than 4.2? Type greater or less.	**greater** (also accepted: greater than, greater than 4.2, bigger)	0.7 is less than 1, so more than 4.2 groups of 0.7 fit. The quotient (6) is greater than 4.2.
t2_s1	Set 2: Which product is LESS than 9? [9 × 0.95 / 9 × 1.05 / 9 × 2.5 / 9 × 1]	**9 × 0.95**	0.95 is less than 1, so 9 × 0.95 is a little less than 9. 9 × 1 equals 9, and the others are greater than 9.

ID	Question	Answer	Why (explain to student)
t1_s2	Set 3: Estimate: round each factor to the nearest whole number. 7.9 × 3.2 is about what number? About:	**24**	7.9 rounds to 8 and 3.2 rounds to 3, so the estimate is 8 × 3 = 24. (The exact product is 25.28.)
t2_s2	Set 3: Leandro says 5.6 ÷ 0.8 = 0.7. Use the prediction rule. Which statement is true? [The quotient must be greater than 5.6, so 0.7 is wrong; it is 7 / 0.7 is right, because dividing makes numbers smaller / The quotient must be less than 0.8, so 0.7 is right / The quotient must be 56]	**The quotient must be greater than 5.6, so 0.7 is wrong; it is 7**	The divisor 0.8 is less than 1, so the quotient is greater than 5.6. Multiply both by 10: 56 ÷ 8 = 7.

REAL WORLD: TIMES AND PACES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation The girls' race is 3 kilometers. Ms. Leighton records Lavinia's time as 13.8 minutes.* Two steps: find Lavinia's pace. If she kept that same pace, how many minutes would a 5-kilometer race take her? Minutes:	**23** (also accepted: 23 minutes, 23 minute, 23 min)	Pace: 13.8 ÷ 3 = 4.6 minutes per kilometer. Five kilometers: 4.6 × 5 = 23 minutes.
w1_s1	Set 2: *The Situation On a long practice day, Leandro runs 2.5 kilometers at a pace of 5.2 minutes per kilometer. Then he walks 0.5 kilometer at a pace of 9.6 minutes per kilometer.* Two steps: how many minutes does the whole practice take? Minutes:	**17.8** (also accepted: 17.8 minutes, 17.8 minute, 17.8 min)	Run: 5.2 × 2.5 = 13 minutes. Walk: 9.6 × 0.5 = 4.8 minutes. Total: 13 + 4.8 = 17.8 minutes.
w1_s2	Set 3: *The Situation Lanelle runs 4 laps of a 0.75-kilometer loop around the meet's practice field. Her watch shows 15.6 minutes in all.* Two steps: what is Lanelle's pace in minutes per kilometer? Minutes per kilometer:	**5.2** (also accepted with a unit word, such as 5.2 minutes or 5.2 minute per kilometer, or 3 other unit spellings)	Distance: 4 × 0.75 = 3 kilometers. Pace: 15.6 ÷ 3 = 5.2 minutes per kilometer.

REAL WORLD: RACING AT ALTITUDE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Boulder is 5,430 feet above sea level, so Coach Larkin reminds runners who live near sea level to drink plenty of water. He asks each visiting runner to drink 0.6 liter of water before the race and 1.5 times that amount after the race.* Two steps: how many liters of water should each visiting runner drink in all? [1.5 liters / 0.9 liter / 2.1 liters / 0.15 liter]	**1.5 liters**	After: 1.5 × 0.6 = 0.9 liter. In all: 0.6 + 0.9 = 1.5 liters.

ID	Question	Answer	Why (explain to student)
w2_s1	Set 2: *The Situation Lanelle has 1.2 kilograms of sports drink powder at the water table. Each jug needs 0.15 kilogram of powder, and each jug holds 3.5 liters of sports drink.* Two steps: if Lanelle uses all of the powder, how many liters of sports drink can she make? Liters:	**28** (also accepted: 28 liters, 28 liter, 28 L)	Jugs: 1.2 ÷ 0.15 = 120 ÷ 15 = 8 jugs. Liters: 8 × 3.5 = 28 liters.
w2_s2	Set 3: *The Situation On the course map, each centimeter stands for 0.4 kilometer. The race route on the map is 7.5 centimeters long. Leandro's practice route is 2.6 kilometers long.* Two steps: how much longer is the real race route than Leandro's practice route? [0.4 kilometer / 3 kilometers / 0.04 kilometer / 4 kilometers]	**0.4 kilometer**	Real route: 7.5 × 0.4 = 3.0 kilometers. Difference: 3.0 − 2.6 = 0.4 kilometer.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct for 0.4 × 0.2 = 0.08? [Tenths times tenths make hundredths: 4 tenths × 2 tenths = 8 hundredths / 4 × 2 = 8, so the answer is 0.8 / Multiply 4 × 2 and keep only one decimal place / The product must be bigger than both factors]	**Tenths times tenths make hundredths: 4 tenths × 2 tenths = 8 hundredths**	A tenth of a tenth is a hundredth, so 4 tenths × 2 tenths = 8 hundredths = 0.08. Count the decimal places: one in each factor makes two.
x2_s0	Set 1: Find the mistake: Leandro says 1.5 × 2.5 = 37.5. What should he fix? [There are two decimal places in all: 15 × 25 = 375, so the product is 3.75 / Nothing. 37.5 is correct / It should be 0.375 / It should be 375]	**There are two decimal places in all: 15 × 25 = 375, so the product is 3.75**	Estimate: 2 × 2.5 = 5, so the answer is near 4, not 37.5. Each factor has one decimal place, so 375 becomes 3.75.
x1_s1	Set 2: Which explanation is correct for 3.6 ÷ 0.9 = 4? [Multiply both numbers by 10: 36 ÷ 9 = 4, and the quotient does not change / Dividing makes a number smaller, so the answer must be less than 3.6 / 0.9 is close to 1, so the quotient is exactly 3.6 ÷ 1 / 0.9 × 4 = 0.36, which matches 3.6]	**Multiply both numbers by 10: 36 ÷ 9 = 4, and the quotient does not change**	Multiplying the dividend and divisor by 10 keeps the quotient the same: 36 ÷ 9 = 4. Check: 0.9 × 4 = 3.6.
x2_s1	Set 2: Find the mistake: Lanelle says 2.4 ÷ 0.6 = 0.4. What should she fix? [Multiply both numbers by 10: 24 ÷ 6 = 4 / Nothing. 0.4 is correct / It should be 40 / It should be 0.04]	**Multiply both numbers by 10: 24 ÷ 6 = 4**	The divisor 0.6 is less than 1, so the quotient must be greater than 2.4. 24 ÷ 6 = 4, and 4 × 0.6 = 2.4.
x1_s2	Set 3: Which explanation is correct? Is 5 × 0.6 more or less than 5? [Less: 0.6 is less than 1, so the product is only part of 5; 5 × 0.6 = 3 / More: multiplying always makes numbers bigger / More: 5 × 6 = 30 / Equal: 0.6 rounds to 1, so the product is 5]	**Less: 0.6 is less than 1, so the product is only part of 5; 5 × 0.6 = 3**	Multiplying by a number less than 1 takes only part of the number. 6 tenths of 5 is 3.

ID	Question	Answer	Why (explain to student)
x2_s2	Set 3: Find the mistake: Coach Larkin's helper estimates 4.8 × 6.1 as 5 × 6 = 30, then writes the exact product as 2.928. What should the helper fix? [The estimate shows the product is about 30, so it is 29.28 / Nothing. 2.928 is correct / It should be 292.8 / It should be 0.2928]	**The estimate shows the product is about 30, so it is 29.28**	48 × 61 = 2,928, and there are two decimal places in all, so the product is 29.28. That is close to the estimate of 30.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: lavinia asked, how long is the race	**Lavinia asked, "How long is the race?"**	Capitalize the name Lavinia, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The meet is in boulder, colorado, near the flatirons.	**The meet is in Boulder, Colorado, near the Flatirons.**	Names of cities, states, and landforms are proper nouns, so capitalize Boulder, Colorado, and Flatirons.
s3_s0	Set 1: Fix this sentence: We packed water snacks and jackets for the meet.	**We packed water, snacks, and jackets for the meet.** (also accepted: We packed water, snacks and jackets for the meet.)	Use commas to separate the items in a series: water, snacks, and jackets.
s4_s0	Set 1: Fix this sentence: each of the runners have a race number	**Each of the runners has a race number.**	Capitalize the first word. The subject is each (one runner at a time), so use has, not have. End with a period.
s1_s1	Set 2: Fix this sentence: coach larkin said, run your own pace	**Coach Larkin said, "Run your own pace."** (also accepted: Coach Larkin said, "Run your own pace!")	Capitalize Coach, which is a title before a name, and the name Larkin. Put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: The runners tied they're shoes before the race.	**The runners tied their shoes before the race.**	Their shows ownership: the shoes belong to the runners. They're means they are.
s3_s1	Set 2: Fix this sentence: Ms. Leighton the head timer started the clock.	**Ms. Leighton, the head timer, started the clock.**	The head timer describes Ms. Leighton. Put a comma before and after a phrase like this that renames the person.
s4_s1	Set 2: Fix this sentence: the times on the scoreboard is correct	**The times on the scoreboard are correct.**	Capitalize the first word. The subject is times (plural), not scoreboard, so use are, not is. End with a period.
s1_s2	Set 3: Fix this sentence: leandro yelled, keep going	**Leandro yelled, "Keep going!"** (also accepted: Leandro yelled, "Keep going.")	Capitalize the name Leandro, put quotation marks around his exact words, capitalize the first word of the quote, and put the end mark inside the closing quotation mark.

ID	Question	Answer	Why (explain to student)
s2_s2	Set 3: Fix this sentence: Lanelle wants to run the hill trail to.	**Lanelle wants to run the hill trail too.**	Too means also: Lanelle wants to run the hill trail, also. The first to (to run) is correct.
s3_s2	Set 3: Fix this sentence: Yes I finished the whole race.	**Yes, I finished the whole race.**	Put a comma after an opening word like Yes, No, or Well.
s4_s2	Set 3: Fix this sentence: the hills on the course is steep	**The hills on the course are steep.**	Capitalize the first word. The subject is hills (plural), not course, so use are, not is. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 0.2 × 0.3	**0.06**	2 points if matched on the first try.
match_s0_2	Set 1: Match 0.5 × 0.5	**0.25**	2 points if matched on the first try.
match_s0_3	Set 1: Match 1.2 × 0.4	**0.48**	2 points if matched on the first try.
match_s0_4	Set 1: Match 2.5 × 1.2	**3**	2 points if matched on the first try.
match_s0_5	Set 1: Match 0.7 × 0.9	**0.63**	2 points if matched on the first try.
match_s1_1	Set 2: Match 2.4 ÷ 0.6	**4**	2 points if matched on the first try.
match_s1_2	Set 2: Match 4.5 ÷ 0.5	**9**	2 points if matched on the first try.
match_s1_3	Set 2: Match 0.56 ÷ 0.08	**7**	2 points if matched on the first try.
match_s1_4	Set 2: Match 1.2 ÷ 0.2	**6**	2 points if matched on the first try.
match_s1_5	Set 2: Match 0.75 ÷ 0.25	**3**	2 points if matched on the first try.
match_s2_1	Set 3: Match the number you divide by	**divisor**	2 points if matched on the first try.
match_s2_2	Set 3: Match the number being divided	**dividend**	2 points if matched on the first try.
match_s2_3	Set 3: Match the answer to a division problem	**quotient**	2 points if matched on the first try.
match_s2_4	Set 3: Match minutes to run one kilometer	**pace**	2 points if matched on the first try.
match_s2_5	Set 3: Match a tenth of a tenth	**hundredth**	2 points if matched on the first try.

BOSS BATTLE: THE CHAMPIONSHIP MEET (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *The Relay Results Lavinia's team runs a relay with 4 runners. Each runner's leg is 1.5 kilometers. Ms. Leighton's clock shows the team's total time: 25.2 minutes.* Which expression gives the team's pace in minutes per kilometer? [25.2 ÷ (4 × 1.5) / 25.2 ÷ 4 × 1.5 / (25.2 × 4) ÷ 1.5 / 25.2 ÷ 1.5 + 4]	**25.2 ÷ (4 × 1.5)**	Pace is total minutes ÷ total kilometers. The total distance is 4 × 1.5, so it goes in parentheses: 25.2 ÷ (4 × 1.5).
b2_s0	Set 1: Two steps: what is the team's pace? Minutes per kilometer:	**4.2** (also accepted with a unit word, such as 4.2 minutes or 4.2 minute per kilometer, or 3 other unit spellings)	Distance: 4 × 1.5 = 6 kilometers. Pace: 25.2 ÷ 6 = 4.2 minutes per kilometer.

ID	Question	Answer	Why (explain to student)
b1_s1	Set 2: *The Orange Slice Sale To raise money, the team buys 2.5 kilograms of oranges at $3.40 per kilogram. They cut the oranges into slices and pack them in bags of 0.25 kilogram each. They sell every bag for $1.50.* How much do the oranges cost? [$8.50 / $85.00 / $0.85 / $5.90]	**$8.50**	2.5 × 3.40: 25 × 34 = 850, with two decimal places in all: $8.50. Estimate: 3 × 3 = 9, so $8.50 makes sense.
b2_s1	Set 2: Two steps: how much money do they collect when they sell all the bags? Dollars:	**15** (also accepted: 15 dollars, 15 dollar, 15.00, 15.00 dollars, 15.00 dollar)	Bags: 2.5 ÷ 0.25 = 250 ÷ 25 = 10 bags. Money: 10 × 1.50 = $15.00.
b1_s2	Set 3: *The Team Sign Leandro paints a rectangular team sign that is 1.8 meters long and 0.6 meter tall. Each small jar of paint covers 0.36 square meter.* What is the area of the sign? [1.08 square meters / 10.8 square meters / 0.108 square meter / 2.4 square meters]	**1.08 square meters**	1.8 × 0.6: 18 × 6 = 108, with two decimal places: 1.08 square meters. (2.4 is the sum, not the product.)
b2_s2	Set 3: Two steps: how many jars of paint does Leandro need to cover the sign exactly once? Jars:	**3** (also accepted: 3 jars, three)	Area: 1.8 × 0.6 = 1.08 square meters. Jars: 1.08 ÷ 0.36 = 108 ÷ 36 = 3 jars.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W28.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: a) 4.6×10^2 b) $53 \div 10^3$ c) 10 × 10 × 10 × 10 × 10 with an exponent	a) 460 b) 0.053 c) 10^5
2	3/10 × 7/10, then as a decimal	21/100 = 0.21
3	Estimate by rounding: a) 3.9 × 2.1 b) 6.2 × 4.8	a) 4 × 2 = 8 b) 6 × 5 = 30
4	How many 0.5-kilometer sections fit in a 3-kilometer course?	6
5	Leandro says 0.4 × 0.2 = 0.8	Model: I disagree. 4 tenths times 2 tenths is 8 hundredths, so 0.4 × 0.2 = 0.08. The product of two numbers less than 1 must be smaller than both of them.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	a) 0.7 × 0.6 b) 2.3 × 1.4 c) 4.5 × 0.8	a) 0.42 b) 3.22 (23 × 14 = 322; two decimal places) c) 3.6 (45 × 8 = 360, so 3.60)
2	a) 5.6 ÷ 0.7 b) 0.81 ÷ 0.09 c) 7.5 ÷ 1.5	a) 56 ÷ 7 = 8 b) 81 ÷ 9 = 9 c) 75 ÷ 15 = 5
3	Lanelle runs 3 kilometers in 16.2 minutes: pace, then 4.5 kilometers	Pace: 16.2 ÷ 3 = 5.4 minutes per kilometer. 4.5 × 5.4 = 24.3 minutes.

#	Prompt	Model answer / grading note
4	Leandro says 3.2 ÷ 0.8 = 0.4	Model: Leandro put the decimal point in the wrong place. Multiplying both numbers by 10 gives 32 ÷ 8 = 4, so the quotient is 4. Dividing by a number less than 1 must give an answer greater than 3.2, and 4 × 0.8 = 3.2 checks it.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Order 7.2 × 0.9, 7.2 ÷ 0.9 and 7.2 without computing; then check	7.2 × 0.9 < 7.2 < 7.2 ÷ 0.9. Multiplying by 0.9 (less than 1) makes a smaller number, and dividing by 0.9 makes a bigger one. Check: 7.2 × 0.9 = 6.48 and 7.2 ÷ 0.9 = 72 ÷ 9 = 8.

Confidential · KingCretot Experience, LLC · PO Box 2322, Arcadia, FL 34265
Approved Provider: Step Up For Students · Arizona ESA · West Virginia Hope Scholarship
EDUCATE. EMPOWER. EXCEL.